

EX:6B	DATA VISUALIZATION USING POWER BI (IMPORTING DATASET)
DATE:	

AIM

To import, transform, analyze, and visualize the Breast Cancer dataset using Power BI Desktop and create interactive dashboards and reports for effective data analysis and decision-making.

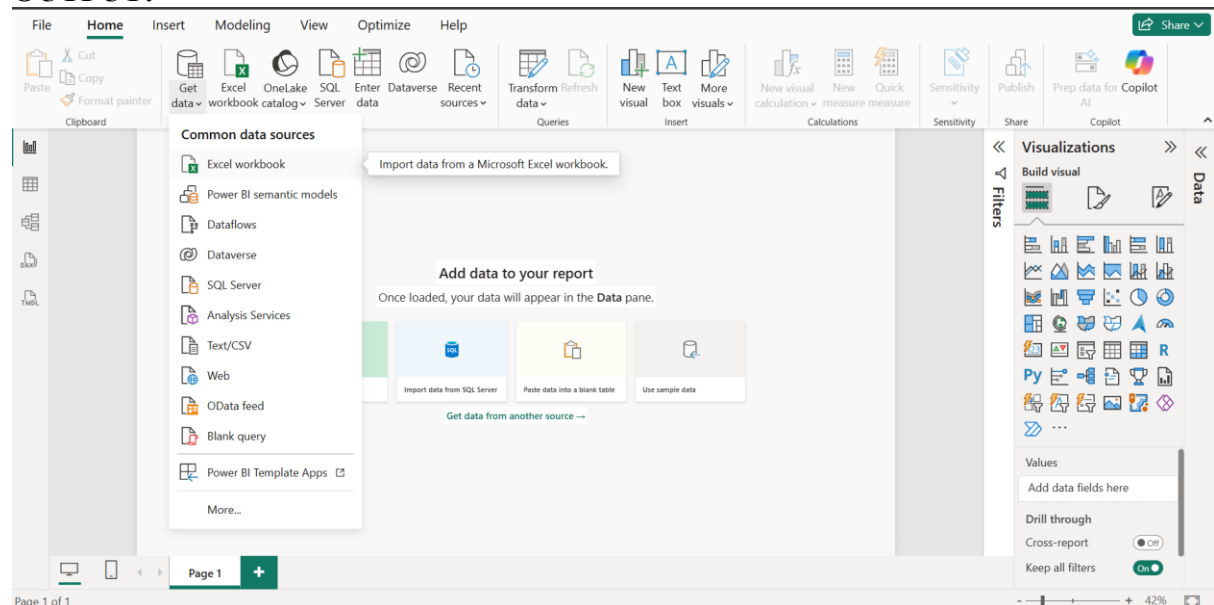
PROGRAM:

1)IMPORT EXCEL DATASET INTO POWER BI

ALGORITHM:

- Step 1: Open Power BI Desktop.
- Step 2: Select Home → Get Data.
- Step 3: Choose Excel Workbook.
- Step 4: Browse and select the Breast Cancer Excel dataset.
- Step 5: Select the required worksheet from the Navigator window.
- Step 6: Select Transform Data to open Power Query Editor.
- Step 7: Verify that the required columns are present.
- Step 8: Proceed with data transformation and cleaning.

OUTPUT:



File Home Help Table tools

Name: breast-cancer

Structure Relationships Calculations Calendars

Manage relationships New measure Quick measure New column New table Mark as date table

Table: breast-cancer (569 rows)

id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	conca
8510426	B	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	
8510653	B	13.08	15.71	85.63	520	0.1075	0.127	0.04568	
8510824	B	9.504	12.44	60.34	273.9	0.1024	0.06492	0.02956	
854941	B	13.03	18.42	82.61	523.8	0.08983	0.03766	0.02562	
85713702	B	8.196	16.84	51.71	201.9	0.086	0.05943	0.01588	
857155	B	12.05	14.63	78.04	449.3	0.1031	0.09092	0.06592	
857156	B	13.49	22.3	86.91	561	0.08752	0.07698	0.04751	
857343	B	11.76	21.6	74.72	427.9	0.08637	0.04966	0.01657	
857373	B	13.64	16.34	87.21	571.8	0.07685	0.06059	0.01857	
857374	B	11.94	18.24	75.71	437.6	0.08261	0.04751	0.01972	
85759902	B	11.52	18.75	73.34	409	0.09524	0.05473	0.03036	
857810	B	13.05	19.31	82.61	527.2	0.0806	0.03789	0.000692	
858477	B	8.618	11.79	54.34	224.5	0.09752	0.05272	0.02061	
858970	B	10.17	14.88	64.55	311.9	0.1134	0.08061	0.01084	
858981	B	8.598	20.98	54.66	221.8	0.1243	0.08963	0.03	
859196	B	9.173	13.86	59.2	260.9	0.07721	0.08751	0.05988	
859464	B	9.465	21.01	60.11	263.4	0.1044	0.07773	0.02172	
859465	B	11.31	19.04	71.8	394.1	0.08139	0.04701	0.03709	
859471	B	9.029	17.33	58.79	250.5	0.1066	0.1413	0.313	
859487	B	12.78	16.49	81.37	502.5	0.09831	0.05234	0.03653	

2) DATA TRANSFORMATION USING POWER QUERY

ALGORITHM:

Step 1: Open Power Query Editor using Transform Data.

Step 2: Examine the imported Breast Cancer table and verify the required columns.

Step 3: Remove blank rows from the dataset.

Step 4: Remove duplicate records.

Step 5: Remove the unnecessary Unnamed: 32 column, if present.

Step 6: Check and handle missing values in the dataset.

Step 7: Set id as Whole Number.

Step 8: Set diagnosis as Text.

Step 9: Set numerical measurement columns such as radius_mean, texture_mean, and area_mean as Decimal Number.

Step 10: Create a derived column named RadiusCategory to classify radius values.

Step 11: Review the Applied Steps pane.

Step 12: Select Home → Close & Apply to load the transformed data.

OUTPUT:

1) Removing blank rows:

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Combine

Queries [1] breast-cancer

Table.TransformColumnTypes(*Promoted Headers,{"id", Int64.Type})

id	diagnosis	radius_mean	texture_mean
842302	M	17.99	
842517	M	20.57	
84300903	M	19.69	21.25
84348301	M	11.42	20.38
84358402	M	20.29	14.34
843786	M	12.45	15.7
844359	M	18.25	19.98
84458202	M	13.71	20.83
844981	M	13	21.82
84501001	M	12.46	24.04
845636	M	16.02	23.24
84610002	M	15.78	17.89
846226	M	19.17	24.8
846381	M	15.85	23.95

32 COLUMNS, 569 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 2008

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type

2) Removing duplicate records

The screenshot shows the Power Query Editor interface. The 'Remove Duplicates' menu option is highlighted under the 'Remove Rows' group. The data table has columns: id, diagnosis, radius_mean, and texture_mean. The 'APPLIED STEPS' pane on the right shows 'Removed Blank Rows' as the last step.

id	diagnosis	radius_mean	texture_mean
842302	M	17.99	10.38
842517	M	20.57	17.77
84300903	M	19.69	21.25
84348301	M	11.42	20.38
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84610002	M	15.78	17.89
846226	M	19.17	24.8
846381	M	15.85	23.95

3) Setting ID as whole number

The screenshot shows the Power Query Editor interface. The 'Changed Type' menu option is highlighted under the 'Transform' group. The data table has columns: id, diagnosis, radius_mean, and texture_mean. The 'APPLIED STEPS' pane on the right shows 'Changed Type' as the last step.

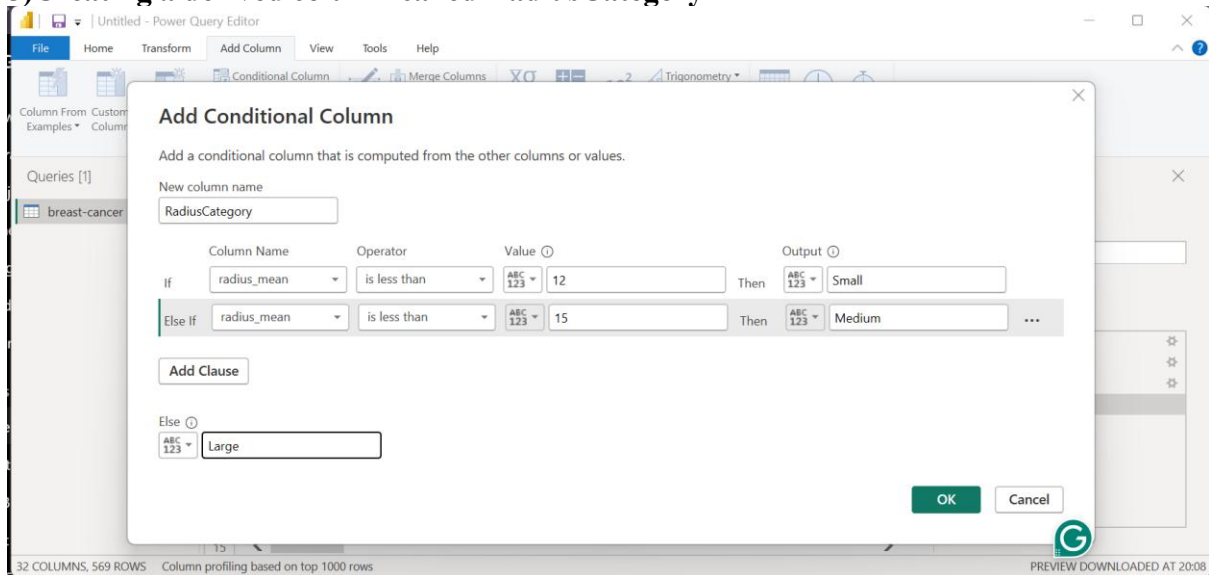
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84501001	M	12.46	24.04
845636	M	16.02	23.24
84610002	M	15.78	17.89
846226	M	19.17	24.8
846381	M	15.85	23.95

4) Set diagnosis as text

The screenshot shows the Power Query Editor interface. The 'Text' data type is selected for the 'diagnosis' column. The data table has columns: id, diagnosis, radius_mean, and texture_mean. The 'APPLIED STEPS' pane on the right shows 'Changed Type' as the last step.

id	diagnosis	radius_mean	texture_mean
842302	M	17.99	10.38
842517	M	20.57	17.77
84300903	M	19.69	21.25
84348301	M	11.42	20.38
84358402	M	20.29	14.34
843786	M	12.45	15.7
844359	M	18.25	19.98
84458202	M	13.71	20.83
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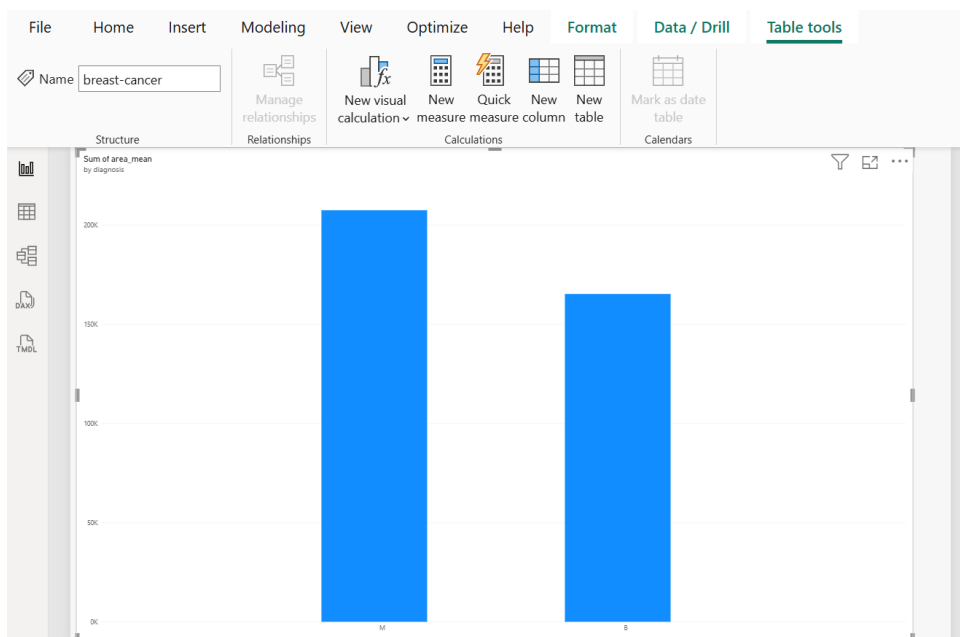
5)Creating a derived column called RadiusCategory



3)VERIFYING THE ETL PROCESS USING A BAR CHART ALGORITHM:

- Step 1: Switch to Report View.
- Step 2: Select Clustered Column Chart.
- Step 3: Drag diagnosis to the X-axis.
- Step 4: Drag area_mean to the Y-axis.
- Step 5: Set the aggregation of area_mean to Average.
- Step 6: Verify that the chart displays the two diagnosis categories: B and M.
- Step 7: Add a suitable title such as Average Tumor Area by Diagnosis.

OUTPUT:



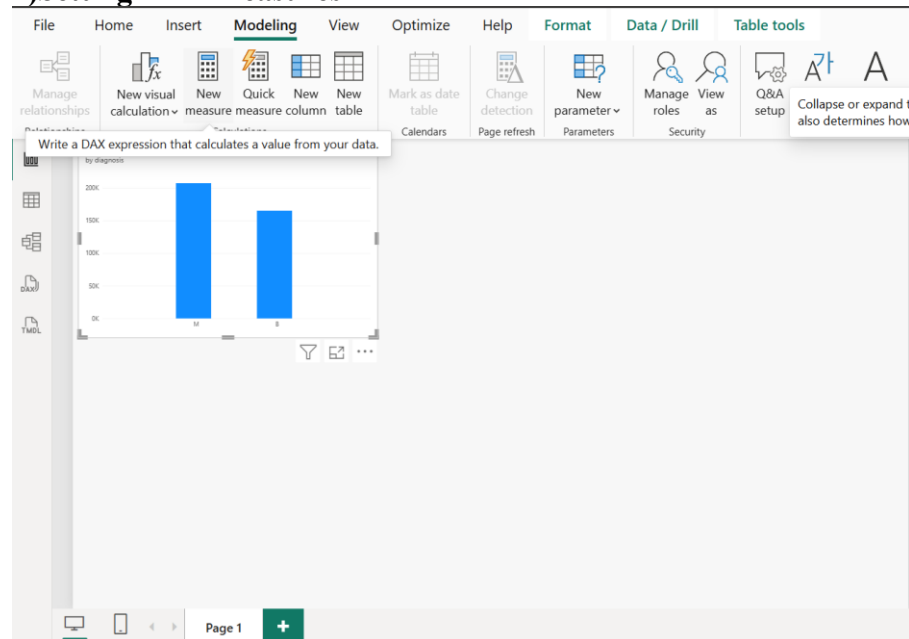
4)CREATING KPI CARDS AND DAX MEASURES:

ALGORITHM:

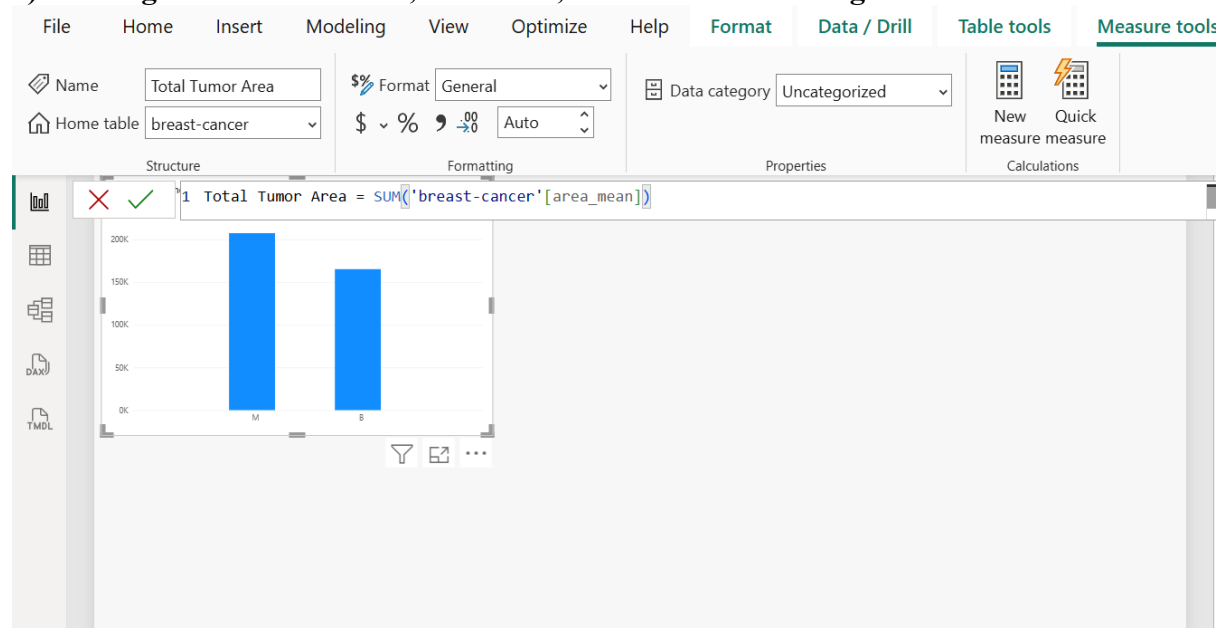
- Step 1: Select the Pie Chart visualization.
- Step 2: Drag Category to Legend.
- Step 3: Drag Sales Amount to Values.
- Step 4: Enable data labels.
- Step 5: Add the title Sales Contribution by Category.
- Step 6: Format the chart.
- Step 7: Display the visualization.

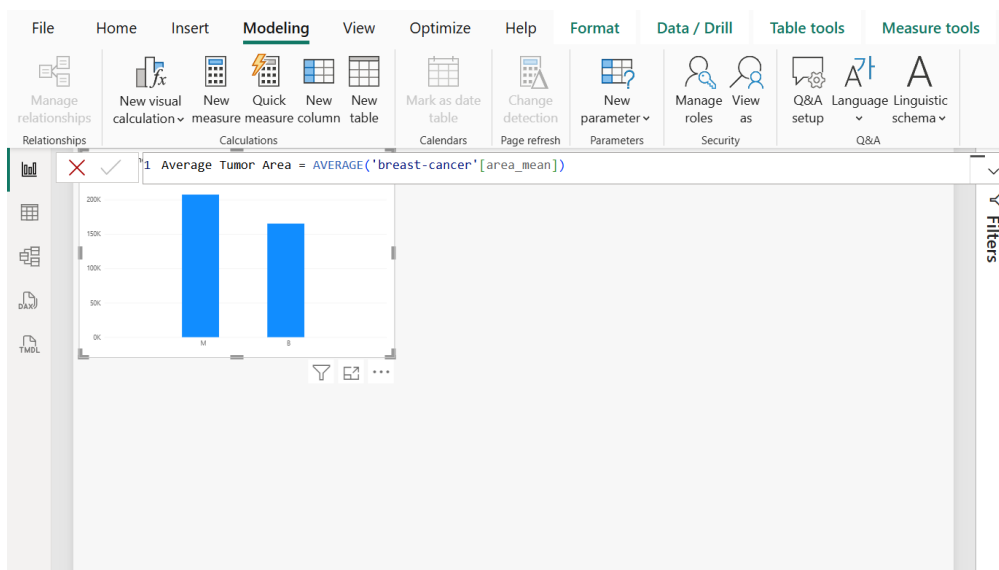
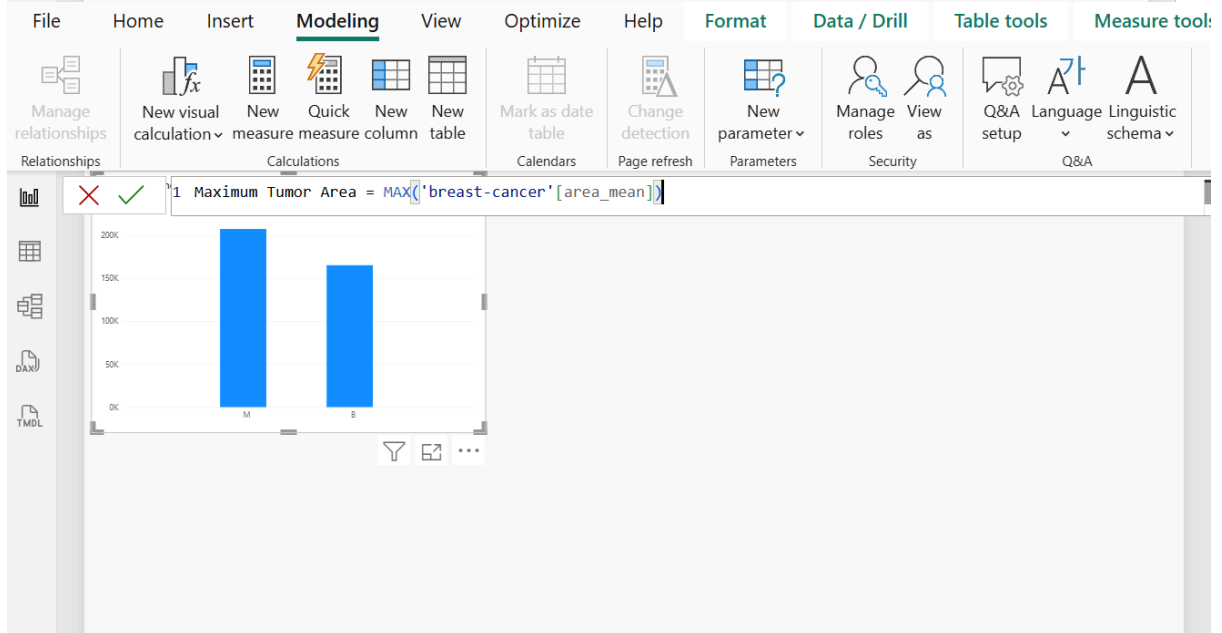
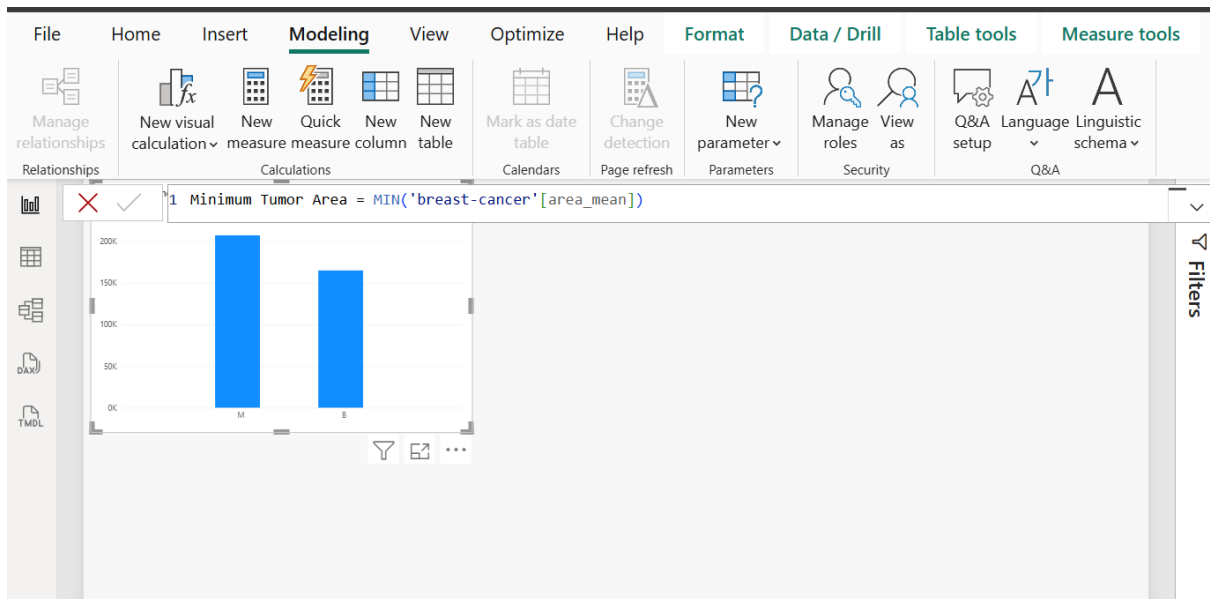
OUTPUT:

1)Setting DAX measures

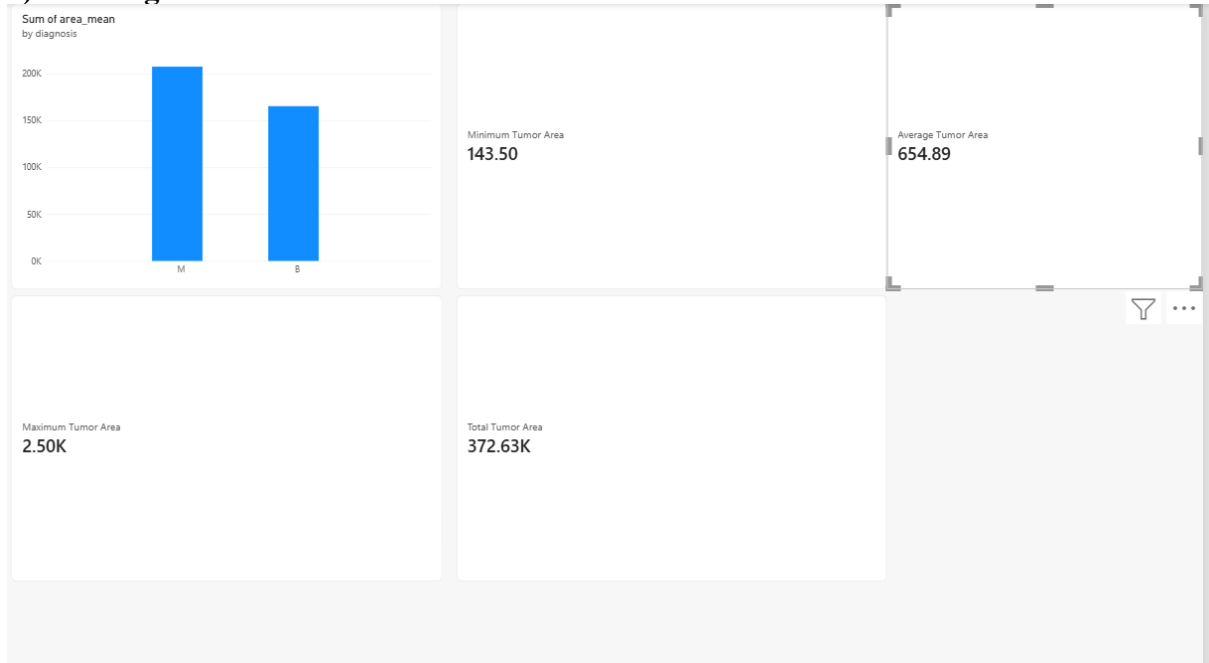


2)Creating the measures total, minimum, maximum and average tumor area:





3) Creating KPI cards

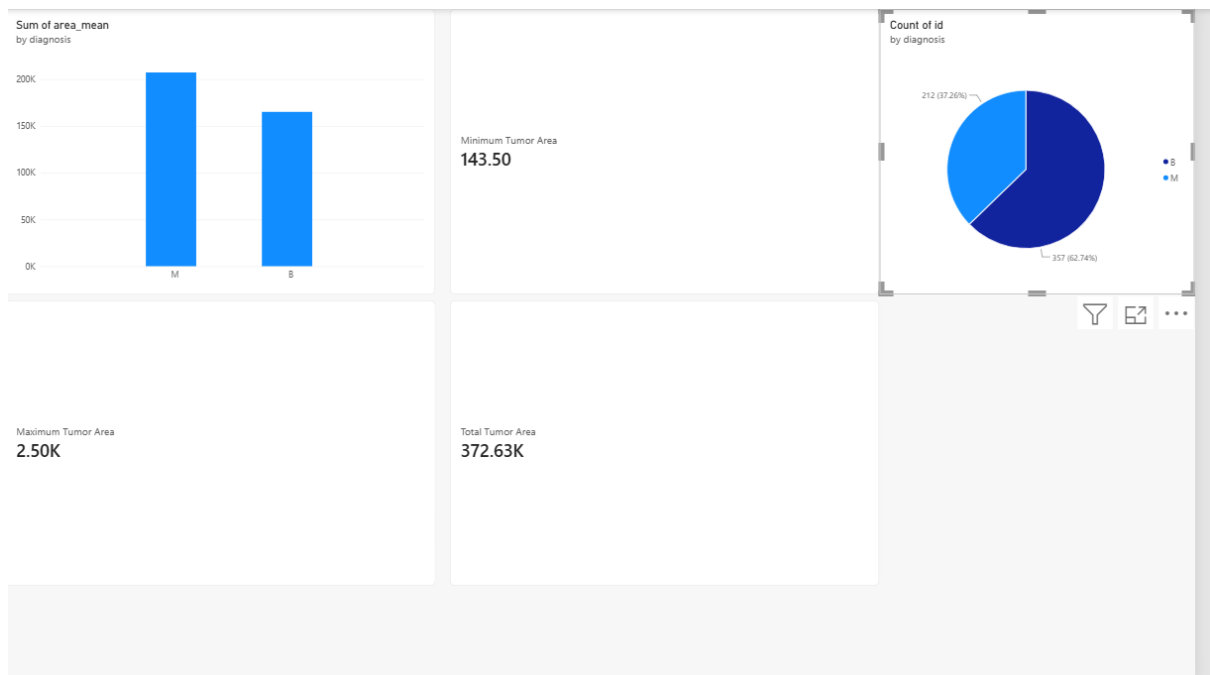


5)PIE CHART – PROPORTION OF BENIGN AND MALIGNANT CASES

ALGORITHM:

- Step 1: Select Pie Chart visualization.
- Step 2: Drag diagnosis to Legend.
- Step 3: Drag id to Values.
- Step 4: Change the aggregation to Count.
- Step 5: Enable data labels/percentages.
- Step 6: Give the title:
Breast Cancer Diagnosis Distribution

OUTPUT:



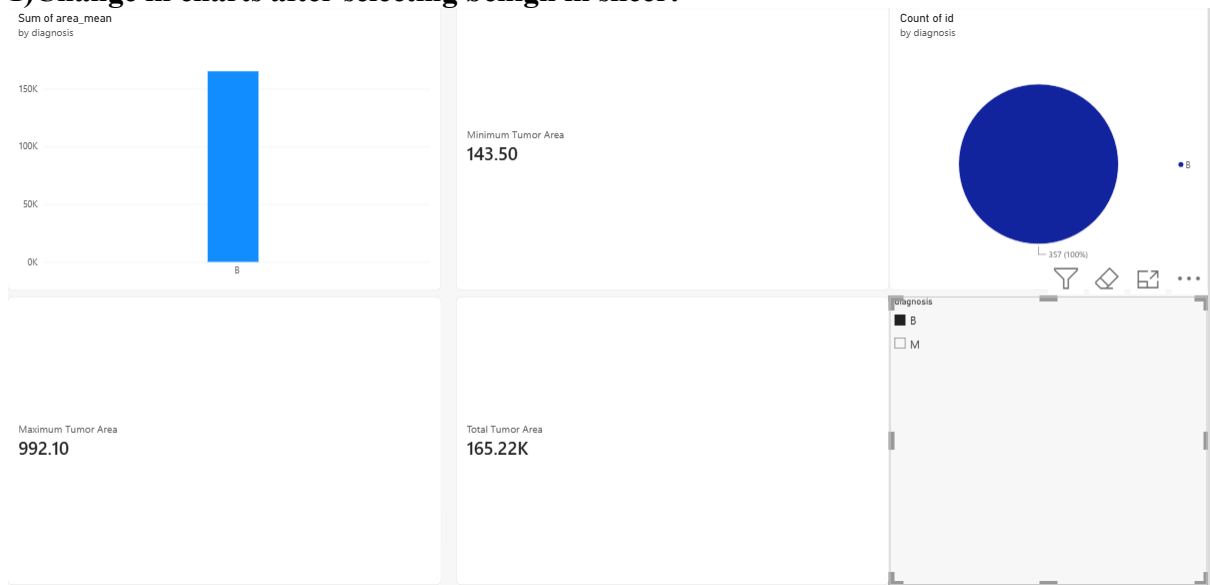
6) CREATING SLICERS

ALGORITHM:

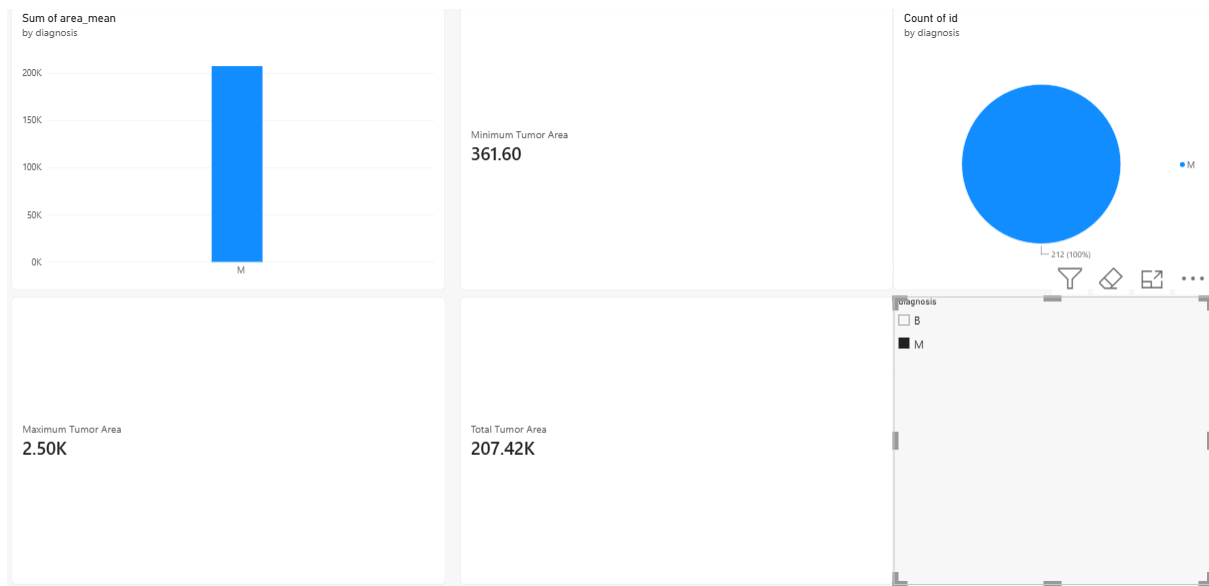
- Step 1: Select the Slicer visualization.
- Step 2: Drag diagnosis into the slicer's Field area.
- Step 3: Enable Single Selection if available.
- Step 4: The slicer will display B and M.
- Step 5: Select B to display only benign cases.
- Step 6: Select M to display only malignant cases.
- Step 7: Observe the changes in the KPI cards and charts.
- Step 8: Clear the selection to return to the complete dataset.

OUTPUT:

1) Change in charts after selecting benign in slicer:



2) Change in charts after selecting malignant in slicer:

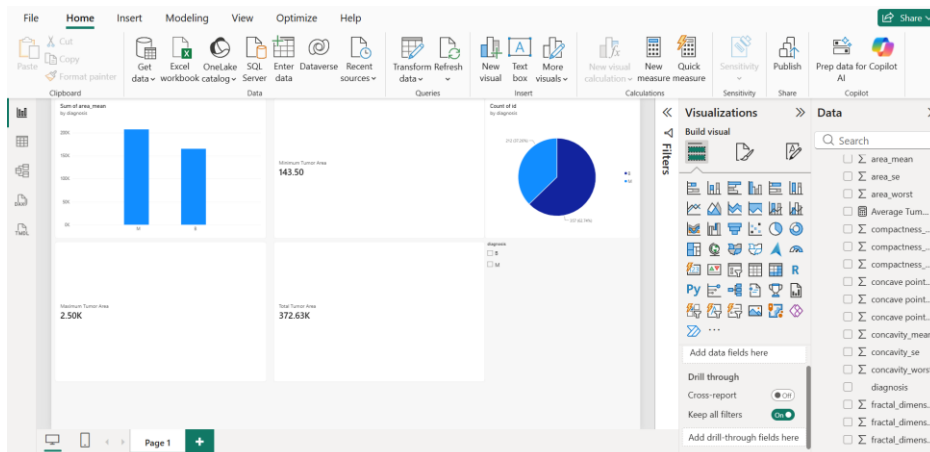


7) CREATING DASHBOARD

ALGORITHM:

- Step 1: Add the KPI Cards.
- Step 2: Add the Bar Chart.
- Step 3: Add the Pie Chart.
- Step 4: Add the Diagnosis Slicer.
- Step 5: Arrange all the visualizations neatly on the report page.
- Step 6: Add a suitable dashboard title.
- Step 7: Apply the diagnosis filter and verify that the visuals respond correctly.
- Step 8: Save the Power BI report.

OUTPUT:



RESULT:

Thus, a Store Sales dataset was successfully created in Microsoft Excel and imported into Power BI. The data was transformed using Power Query, and bar chart, pie chart, KPI cards, DAX measures, and slicers were successfully created. Finally, an interactive Store Sales Analysis Dashboard was developed to visualize and analyze sales data effectively.